

FL-01: Master Workflow Audit, AI Setup & Case Framing Document

Repository: flyrank-ml

Path: work/FL-01_Workflow_Audit.pdf

Track: General AI Fluency & Machine Learning Foundations

Section A: Personal Workflow Audit

#	Recurring Task	Classification	One-Line Rationale
1	Handcrafting organic shampoo batches		

| Just Me

| Physical, artisan-level compounding done in small batches by my mother that requires tactile quality control.

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| 2 | Negotiating raw ingredient prices with local vendors

| Just Me

| Requires building personal trust, navigating local wholesale markets, and real-time human negotiation.

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| 3 | Testing and debugging physical hardware circuits

| Just Me

| Requires physical laboratory instrumentation, multimeters, and manual wiring that AI cannot physically interact with.

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| 4 | Writing complex SQL & DuckDB queries

| Collaborate with AI

| I provide the database schema and query intent; the AI rapidly drafts the initial syntax,

which I then optimize for performance.

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| **5** | Drafting social copy & catalog descriptions for @Khaam.essentials

| **Collaborate with AI**

| I provide the product benefits; the AI refines the tone and structures the copy, which I then localize for our target audience.

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| **6** | Translating numerical analysis algorithms into Python scripts

| **Collaborate with AI**

| Great for translating step-by-step mathematical logic (like Jacobi or Gauss-Seidel) into structured, clean code.

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| **7** | Synthesizing engineering handbook chapters into review sheets

| **Delegate to AI with Review**

| I upload the raw PDF chapter; the AI extracts key formulas and definitions, which I audit against my lecture notes.

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| **8** | Reviewing documentation on rule-based recommendation scoring

| **Delegate to AI with Review**

| The AI summarizes dense API/framework docs so I can quickly extract implementation patterns for my internship tasks.

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| **9** | Triaging basic customer FAQs on Instagram & WhatsApp

| **Delegate to AI with Review**

| Standard questions about shipping, product availability, or price are drafted by AI templates, requiring only my manual click to send.

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| **10** | Formatting, linting, and cleaning local Python code bases

| **Fully Automate**

| Easily managed using automated Git hooks, Black, and Flake8 formatters running locally on file save.

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| **11** | Syncing raw inventory lists with spreadsheet stock counts

| **Fully Automate**

| Handled by simple Google Sheets script triggers that auto-update stock levels whenever a WhatsApp order form is logged.

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| **12** | Generating mock technical interview questions

| **Collaborate with AI**

| Helps me practice core engineering and discrete modeling concepts through interactive chat sessions.

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Section B: AI Setup Evidence & Custom Instructions

Environment Status

- **Claude.ai / ChatGPT Accounts:** Confirmed Active.
- **Anthropic Academy:** Enrolled in Claude 101 (Module 1 Completed).

Custom Instructions Configured

Professional Summary

Technical and detail-oriented Engineering and Computer Science student with proven e backend optimization, numerical simulation, and data-driven systems. Currently leverag SQL, and DuckDB to develop high-performance recommendation engines and automa Strong background in systematic validation, functional testing, and rigorous analytical re

(Insert Screenshot: docs/assets/claude_setup_evidence.png)

Section C: Portfolio Sitemap & Hero Positioning

[Home Page / Hero]

Hero Title: "I build leakage-proof ML pipelines and high-throughput data tools that
Sub-Headline: "From client-grouped validation pipelines that hold 84% Precision@

Primary CTA: "View My Systems & Case Studies"

[Case Study 1: FlyRank Capstone]

"Client-Grouped Validation & Content Decay Prediction (84% Precision@50)"

[Case Study 2: CollabIn Internship]

"Optimizing Recommendation Scoring Loops via DuckDB (70x Speedup)"

[Founder Build: Khaam Essentials]

"Automating E-Commerce Inventory Sync via Python & Google Sheets"

[About & Contact Page]

Technical Bio, Resume Download, & Direct Scheduling Link

Section D: Selected Tasks & Success Metrics

Task 1: Refactoring & Optimizing Recommendation Algorithms

- **Execution Speed:** The optimized DuckDB/Pandas Python script must execute over a target dataset within defined latency bounds.
- **Rigorous Data Separation:** The pipeline must feature strict index partitioning to ensure zero target or feature leakage between training splits.
- **Maintainability:** Code must score 10/10 on structural linting (Flake8/Pylint) and contain descriptive docstrings for team handoffs.

Task 2: Writing Copy for Khaam Essentials

- **Brand Alignment:** Avoids robotic, overly generic AI buzzwords (e.g., *"Revolutionize your hair game!"* or *"Delve into"*).
- **Authenticity:** Masters a natural, storytelling tone that explicitly highlights our core brand pillars: organic, chemical-free, and lovingly handcrafted by my mother.
- **Formatting Constraints:** Copy must adhere to strict platform boundaries (under 150 characters for Instagram Bios or structured bullet points for WhatsApp catalogs).

Task 3: Building Engineering Course Study & Problem Guides

- **Mathematical Accuracy:** Generated practice problems for numerical analysis (root-finding, interpolation) or instrumentation process control must contain 100% correct, verifiable step-by-step solutions.
 - **Syllabus Mapping:** The study guides must map perfectly to university course handbook chapters without introducing irrelevant, out-of-scope concepts.
 - **Clarity:** Formulas must be cleanly formatted using clear Markdown tables and standard notation.
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Section E: Verified Case Studies & Positioning Statements

1. Voice Card

"Direct, metric-first, honest, plain, zero buzzwords."

2. One-Paragraph Proof Statement

I design leak-free validation splits for client-grouped and tabular datasets. I am proving this to a Lead Machine Learning Engineer hiring for a fast-moving ML team who needs developers capable of building reliable evaluation frameworks without hand-holding. The single action I want them to take after reviewing my portfolio is to reply to my message and approve me for a technical screen. I prove my claim by showing how I designed a strict client-grouped split that prevents target leakage and holds an 84.0% Precision@50 on entirely unseen domains.

3. One-Line "Why"

This portfolio exists because a standard resume only lists tools and courses, failing to prove to a skeptical Lead ML Engineer that I possess the engineering judgment to design mathematically rigorous, leakage-free validation splits that hold up on unseen production data.

4. Before / After Fluff Comparison

- **Generic AI Copy (Before):**
"I leveraged advanced machine learning algorithms and robust data processing

frameworks to deliver data-driven insights and optimize content refresh cycles for elevated organic growth."

- **Edited Copy (After):**

"I trained a Random Forest classifier on a strict client-grouped split across 32 domains, achieving an 84% Precision@50 to stop wasting content rewrite budgets on healthy web pages."

5. Case Study 1 (FlyRank Capstone)

Title: 84% Precision@50, Zero Leakage: A Client-Grouped Validation Pipeline for Content Decay

- **Beat 1: The Problem**

Content operations managers had 50 monthly refresh slots and used a static 100-point heuristic (neglect, ranking position, 90-day impressions) to assign them. That heuristic scored ~35% Precision@50 on held-out domains—meaning roughly 33 of every 50 slots (65% of team effort) were spent rewriting pages that were not actually decaying.

- **Beat 2: What I Did (and Decided)**

I framed content decay as a binary classification task (`is_decayed`). Instead of standard random train/test splits that leak sitewide signals (domain authority, backlink profiles) across splits, I implemented a strict client-grouped split (GroupKFold by domain). I trained a Random Forest classifier on multi-factor features, evaluating strictly on unseen domain holdouts. I chose Precision@50 because human writing bandwidth is capped at 50 rewrites per month—optimizing for precision directly protects the creative writing budget from false positives.

- **Beat 3: What Came Of It**

The pipeline achieved 84.0% Precision@50 on 100% unseen domain holdouts, compressing wasted slots from 33 down to just 8 per monthly cycle—a verified 2.4x efficiency lift over the baseline heuristic. The output serves as a validated offline benchmark and leak-free split-design methodology for content-decay prediction—a reusable blueprint for future production engineering.

6. Case Study 2 (CollabIn Internship)

Title: 70x Faster Local Iteration: Optimizing Recommendation Scoring Loops via DuckDB

- **Beat 1: The Problem**

Our backend squad needed a faster way to iterate on multi-factor recommendation scoring weights over a 1.8 GB user-interaction dataset.

Processing this 4.2-million-row matrix using traditional in-memory Pandas pipelines required 84.3 seconds per execution loop and pushed local RAM limits to the brink of crashing.

- **Beat 2: What I Did (and Decided)**

I independently refactored the scoring engine using DuckDB and vectorised SQL instead of row-by-row Pandas `.apply()` lambdas. By executing multi-factor weight calculations natively within a column-oriented database architecture, I kept memory usage negligible while executing directly over the raw dataset.

- **Beat 3: What Came Of It**

Query execution times dropped from 84.3 seconds down to 1.2 seconds—a 70x performance speedup. It continues to replace the legacy Pandas loop for the squad's local testing, maintaining a 1.2-second iteration cycle for backend engineering calibration.

7. Founder Build Callout (Khaam Essentials)

- **Title:** Automation Under Ambiguity: Zero-Spec Inventory Sync
 - **Summary:** Designed and executed Python scripts connecting WhatsApp order inputs directly to live stock sheets, automating local e-commerce operations for my family business without paid software tools.
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Section F: AI Audit Verification & Sign-Off Checklist

(Insert Screenshot: `docs/assets/claude_audit_signoff.png`)

Summary of Audit Fixes (The 9 Caught Fluff Patterns)

1. **"50% of budgets"** → Unsourced assertion; **Cut.**
 2. **"2.4x lift"** → Verified experiment (84% vs. 35% Precision@50); **Kept.**
 3. **"memory overloads"** → Softened to exact technical reality ("brink of crashing"); **Corrected.**
 4. **"in real time"** → Decorative buzzword; **Cut.**
 5. **"prevents catastrophic model decay in production"** → Claimed unmeasured outcome; **Cut.**
 6. **"~95% fake metric"** → Unlogged estimate for random split; **Caught and removed before shipping.**
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7. **"ready for production deployment"** → Implied unconfirmed roadmap status;
Softened to "reusable blueprint".
 8. **"permanently"** → Overclaimed future durability; **Swapped for present-tense fact ("continues to replace")**.
 9. **"site-auditing engine"** → Invented internal product designation; **Cut**.
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Status: Audit Signed Off & Ready for Repository Deployment.